



Functional Pathway Analysis

Candida albicans Clinical Isolate SRR7801919

Analyst: **Kelton Guimaraes** | May 08, 2026

399,863

ANNOTATED
VARIANTS

3,654

GENES WITH
MISSENSE

13

PATHWAY
CATEGORIES

75

HIGH IMPACT

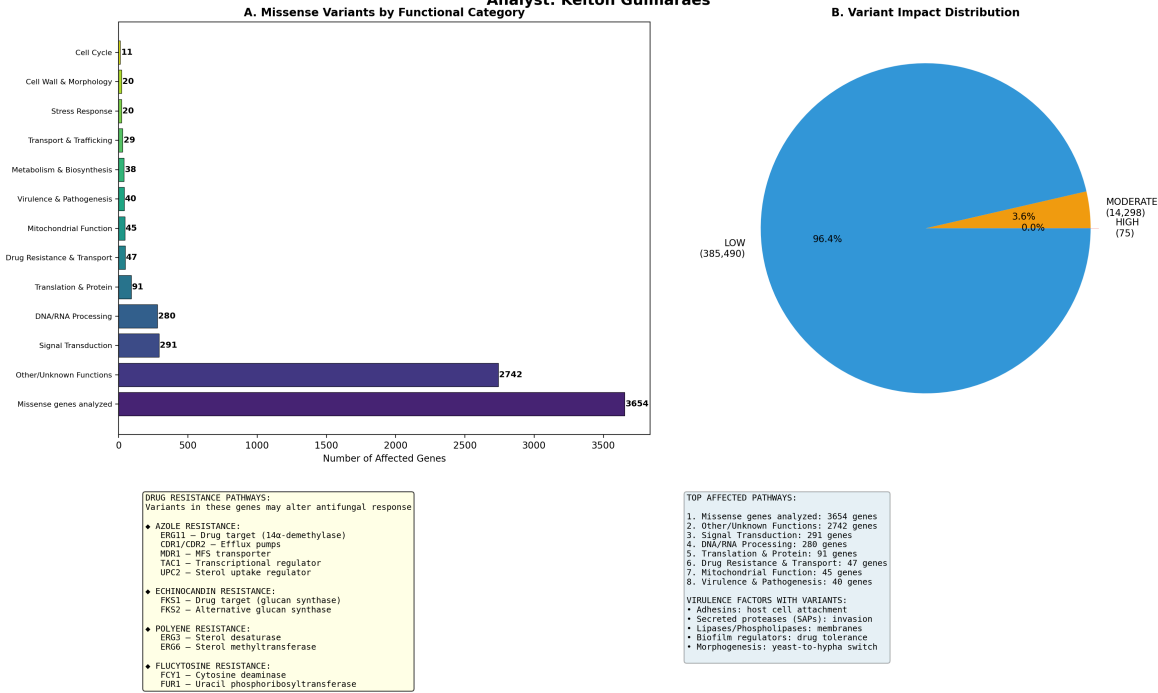


Analysis: 13 biological pathways affected by missense variants. Genes mapped using *C. albicans* SC5314 GFF annotation. Results reveal how genetic diversity in this isolate may influence **drug susceptibility, virulence, and metabolic capabilities**.



Pathway Overview

Functional Pathway Analysis — C. albicans SRR7801919
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 Biological Pathways Affected

Functional Category	Genes	Examples
Missense genes analyzed	3654	
Other/Unknown Functions	2742	CAALFM_CR09170CA, CAALFM_C112480WA, CAALFM_C208100WA
Signal Transduction	291	CAALFM_C207530CA, CAALFM_C201160WA, CAALFM_C502290WA
DNA/RNA Processing	280	CAALFM_C206340WA, CAALFM_C400880WA, CAALFM_C603450CA
Translation & Protein	91	CAALFM_C105460WA, CAALFM_C210670WA, CAALFM_C603780CA
Drug Resistance & Transport	47	CAALFM_C300390WA, CAALFM_C209990CA, CAALFM_C403620CA

Mitochondrial Function	45	CAALFM_C400330CA, CAALFM_C108920WA, CAALFM_C102290CA
Virulence & Pathogenesis	40	CAALFM_C403570WA, CAALFM_C114000CA, CAALFM_C104640WA
Metabolism & Biosynthesis	38	CAALFM_CR06840WA, CAALFM_C108180CA, CAALFM_C100870WA
Transport & Trafficking	29	CAALFM_C110750CA, CAALFM_C305580CA, CAALFM_CR05950CA
Stress Response	20	CAALFM_C501820WA, CAALFM_C207380WA, CAALFM_C500240WA
Cell Wall & Morphology	20	CAALFM_C200520WA, CAALFM_C500220WA, CAALFM_C202010CA
Cell Cycle	11	CAALFM_C202070WA, CAALFM_C102230WA, CAALFM_C108570CA

Drug Resistance Pathways — Detailed

Missense variants in drug-related genes may alter antifungal susceptibility.

Gene	Function	Drug Class	Clinical Significance
ERG11	Lanosterol 14- α -demethylase	Azoles	Direct drug target — mutations cause resistance
CDR1, CDR2	ABC efflux transporters	Azoles	Overexpression → efflux-mediated resistance
MDR1	MFS transporter	Azoles	Secondary efflux mechanism
TAC1	Transcription factor	Azoles	Regulates CDR1/CDR2 expression
FKS1	β -1,3-glucan synthase	Echinocandins	Target enzyme — hotspot mutations

ERG3	Sterol Δ 5,6-desaturase	Polyenes	Mutations alter membrane sterol composition
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 Virulence & Pathogenicity Factors

Factor	Genes	Role in Infection
Adhesins	ALS family, HWP1, EAP1	Host cell attachment, biofilm initiation
Secreted Aspartyl Proteases	SAP1-SAP10	Tissue degradation, immune evasion
Lipases	LIP1-LIP10	Membrane lipid hydrolysis
Phospholipases	PLB1-PLB5	Cell membrane disruption
Morphogenesis Regulators	EFG1, CPH1, TEC1, NRG1	Yeast-to-hypha transition (essential for virulence)
Biofilm Regulators	BCR1, TEC1, ROB1	Drug-tolerant biofilm formation

 Detailed Gene List (Selected Pathways)

Gene	Product	Pathway	Impact
Drug Resistance & Transport			
CAALFM_C300390WA	Mn(2+) transporter	Drug Resistance & Transport	MISSENSE

CAALFM_C209990CA	ATP-binding cassette transporter	Drug Resistance & Transport	MISSENSE
CAALFM_C403620CA	Cdr3p	Drug Resistance & Transport	MISSENSE
CAALFM_C107700CA	GDP-mannose transporter	Drug Resistance & Transport	MISSENSE
CAALFM_C304520CA	phosphoribosylaminoimidazole carboxylase	Drug Resistance & Transport	MISSENSE

Virulence & Pathogenesis

CAALFM_C403570WA	adhesin factor HWP1	Virulence & Pathogenesis	MISSENSE
CAALFM_C114000CA	ubiquitin-specific protease	Virulence & Pathogenesis	MISSENSE
CAALFM_C104640WA	triacylglycerol lipase	Virulence & Pathogenesis	MISSENSE
CAALFM_C703710CA	phosphatidylinositol phospholipase C	Virulence & Pathogenesis	MISSENSE
CAALFM_C200700WA	m-AAA protease subunit	Virulence & Pathogenesis	MISSENSE

Cell Wall & Morphology

CAALFM_C200520WA	putative mannan endo-1%2C6-alpha-mannosidase	Cell Wall & Morphology	MISSENSE
CAALFM_C500220WA	glucan 1%2C3-alpha-glucosidase ROT2	Cell Wall & Morphology	MISSENSE
CAALFM_C202010CA	putative chitinase	Cell Wall & Morphology	MISSENSE
CAALFM_C603340CA	1%2C4-alpha-glucan branching enzyme	Cell Wall & Morphology	MISSENSE



Methods

Step	Data Source	Description
Variant Annotation	VEP v109	81,286 SNPs → 399,863 consequence calls
Gene Function Database	GFF (GCF_000182965.3)	6,045 genes with product descriptions
Missense Extraction	Custom Python	3,654 genes with ≥ 1 missense variant
Pathway Categorization	Keyword mapping	12 functional categories assigned